Percentage & Profit and Loss.

Date 10/03/25

Assignment No 1.

1 25% of 200.

$\frac{25}{\cancel{100}} \times 2\cancel{00}$ $\boxed{= 50\%}$

2. 40% of number is 80.

$\frac{4\cancel{0}}{10\cancel{0}} \times \cancel{80}\, x = 80.$

$x = \frac{\cancel{800}}{\cancel{4}}$ 200.

$\boxed{x = 200}$

3. 75% of num is 150.

$\frac{75}{100} \times x = 150$

$x = \frac{150 \times 100}{75}$

$= \frac{\cancel{15000}}{\cancel{75}}$ 200, $\cancel{600}$, 3, 1

$x = 200.$

Date
Page

4. 15% of 120

$$\frac{\cancel{15}^{3}}{\cancel{100}_{5}} \times \cancel{120}^{6} = 18.$$

5. 30% of num is 90.

$$x \times \frac{3\cancel{0}}{10\cancel{0}} = 90.$$

$$x = \frac{\cancel{900}^{300}}{\cancel{3}}$$

$$x = 300$$

6. price increases from 200 to 250.

$$\uparrow = 200 \longrightarrow 250 \Rightarrow 50.$$

$$\% \uparrow = \frac{50}{20\cancel{0}} \times 10\cancel{0}$$

$$= 25\%$$

7. salary ~~in~~ ↑se from 40000 to 50000 what is the % increase.

$$40000 \rightarrow 50000 \Rightarrow 10000.$$

$$\frac{\cancel{10000}^{25}}{\cancel{40000}} \times 1\cancel{00}.$$

$$\boxed{= 25\%}$$

8. Populatⁿ of town decresed from 10000 to 8000. what is the % ↓.

→

$$10000 \rightarrow 8000.$$

$$\downarrow = 10000 - 8000 = 2000$$

$$= \frac{2\cancel{000}}{1\cancel{0000}} \times \cancel{100} = \boxed{20\%}$$

9. Book price drop from 500 to 400. what is %↓

→

$$500 - 400 = 100.$$

$$\frac{\cancel{100}^{20}}{\cancel{500}} \times \cancel{100} = \boxed{20\%}$$

10. if CP of item is ₹600 & SP = 450. what is % loss ?

→

$$= \frac{600 - 450}{60\cancel{0}} \times 100.$$

$$= \frac{\cancel{150}^{25}}{\cancel{600}} \times \cancel{100}$$

$$\boxed{= 25\%}$$

11. which is greater. 30% of 400 or 40% of 300?

→

$\frac{30}{\cancel{100}} \times 4\cancel{00}$ $\frac{40}{\cancel{100}} \times 3\cancel{00}$

= 120. 120

= both are equal.

12. A person spends 60% of his income & saves ₹ 8000 what is his total income.

→ saves = 8000.

40% = 8000.

80% = 16000.

+20% = 4000

20,000

13. if A is 20% more than. B. B is how much less than A?

→ B = 100.

A = 120.

$\frac{\cancel{20}}{\cancel{120}_{2}} \times 1\cancel{0}\cancel{0} =$

$\frac{20}{12} \times 10 = \frac{\cancel{208}}{12} \overset{16.67}{} \; 80$

$\boxed{= 16.67}$

14. if the price of sugar is increased by 25% by how much should the consumption be reduced to maintain the same expense ?

→

$100 \longrightarrow 125\%$

$$\frac{\cancel{25}^{1}}{\cancel{125}_{5}} \times \cancel{100}^{20}$$

$= 20\%$

15. IF A's. is 40% more than B's income, then B's income is what % less than A's ?

A ⇒ 140
B => 100

$$\frac{40}{14\cancel{0}} \times 10\cancel{0} = \frac{\cancel{400}^{28.5}}{\cancel{14}_{1}}$$

$= 28.57$ ~~40~~

120 ÷ 12 80

16.

→

$100 \longrightarrow 20\% \uparrow \longrightarrow 120 \xrightarrow{10\% \downarrow}$ ~~8% dees~~ 112.

$\boxed{8\% \text{ decrease}}$

$\frac{1\cancel{0}}{1\cancel{0}\cancel{0}} \times 12\cancel{0}$

17

$100 \xrightarrow[\uparrow]{30\%} 130 \xrightarrow[\downarrow]{20\%} 104.$

(4% decrease)

$\frac{2\cancel{0}}{1\cancel{0}\cancel{0}} \times 13\cancel{0} = 26 = \begin{array}{r} 130 \\ -26 \\ \hline 104. \end{array}$

18.

→ $100 \xrightarrow[25\%]{\uparrow} 125 \xrightarrow[20\%]{\downarrow} 100$

(0% change)

$\frac{2\cancel{0}}{10\cancel{0}} \times 125$

$\frac{25\cancel{0}}{1\cancel{0}}$

19. $100 \xrightarrow[\uparrow se]{40\%} 140 \xrightarrow[\downarrow se]{30\%} 98\%$. $\boxed{2\% \text{ decrease}}$

$$\frac{3\cancel{0}}{1\cancel{0}\cancel{0}} \times 14\cancel{0}$$

42

20.

$100 \xrightarrow[20\%]{\uparrow} 120 \xrightarrow[10\%]{\downarrow} 112$

12%.

$$\frac{1\cancel{0}}{10\cancel{0}} \times 12\cancel{0}$$

21. % profit = 25%.

CP = 100

SP = ~~100 + 25%~~.

$$\frac{25}{\cancel{100}} \times \cancel{100}$$

$\boxed{SP = \cancel{100}\ 125}$

22.

→

10% on marked price.

$\frac{10}{\cancel{100}} \times 5\cancel{00} = 50.$ $\frac{18}{1\cancel{00}} \times 5\cancel{00}$

9.0 +8

$$\begin{array}{r} 500 \\ -\ 90 \\ \hline 410. \end{array}$$

$\frac{10}{10\cancel{0}} \times 5\cancel{0}0 = 50. = 450₹$

$\frac{8}{10\cancel{0}} \times \overset{4}{45}\cancel{0}$

$$\begin{array}{r} 450 \\ -\ 36 \\ \hline 414 \end{array}$$

36.0

= 414.

23. CP = 100.

SP = 120.

prof

$$\frac{\cancel{100}^{20}}{120} \times 70\cancel{0} = x.$$

$$\frac{\cancel{1000}^{83}}{\cancel{12}} \quad \frac{20}{12\cancel{0}} \times 1\cancel{00}$$

$$\frac{\cancel{200}^{16.67}}{\cancel{12}} \quad 80 \quad 72.80$$

16.67%.

24.

→ CP = 1200.

SP = 960.

1200 → 960.

$$\frac{\cancel{240}^{20}}{\cancel{1200}} \times 1\cancel{00}$$

$\boxed{= 20\%}$

25. 500 ⟶ ~~600~~. 650.

$\frac{\cancel{150}^{30}}{\cancel{500}} \times 10\cancel{0}$

$\boxed{30\%}$

26.

A = 120.

B = 100

$\frac{\cancel{20}}{\cancel{120}} \times 10\cancel{0} =$

16.66....

$\boxed{= 16.67}$.

27.

B : G = 3 : 2.

$\frac{3}{2} =$ 5 = 3 B & 2 G.

Boys % = $\frac{3}{\cancel{5}} \times \cancel{100}^{20}$

$\boxed{= 60\%}$

28.

→ $2,00000 \longrightarrow 2,50,000$

$$\frac{5\cancel{0,000}}{2\cancel{00,000}} \times \cancel{100}$$

$$\boxed{= 25\%}$$

29.

$65\% - 35\% = 30\% = 3000$ Votes.

$$= \frac{3000 \times 100}{30}$$

$$\frac{10000}{\cancel{300000}}$$

$$\boxed{= 10,000}$$

30.

$100 \xrightarrow[30\%]{} 70$

$$\frac{30}{7\cancel{0}} \times 10\cancel{0}$$

$$\frac{300}{7} = 42.857$$

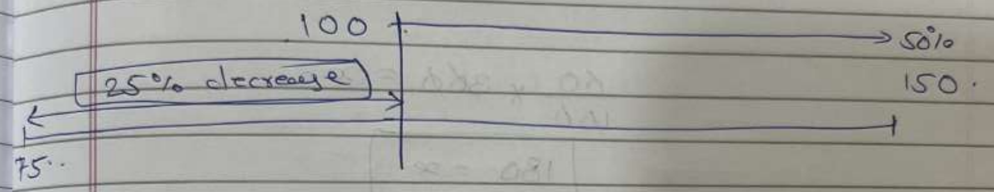
60 40 50

$$\boxed{= 42.85}$$

31.

→ 100 → 150 → 105.

$\frac{\cancel{50}}{\cancel{100}} \times \cancel{150}$



$\boxed{25\ \% \downarrow se}$.

32.

A = ~~100~~ 120

B = 100.

$\frac{20}{12\cancel{0}} \times 10\cancel{0}$

$\frac{\cancel{200}}{\cancel{12}}$ 16.67 6

$\boxed{= 16.67\ \%}$

33.

$$\frac{80}{100} \times x = 90$$

$$x = \frac{\cancel{9000}^{300}}{\cancel{30}}$$

$$\boxed{x = 300}$$

$$\frac{60}{100} \times x = \cancel{90}^{300}\ x$$

$$x = \frac{\cancel{9000}^{150}}{\cancel{60}}$$

$$\boxed{x = 150}$$

$$\frac{60}{1\cancel{00}} \times 3\cancel{60} = x$$

$$\boxed{180 = x}$$

$$\boxed{x = 180}$$

34.

→ 25% = 5000.

$$\begin{aligned} 75\% &= 15000 \\ + \quad 25\% &= 5000 \\ \hline & \boxed{20{,}000} \end{aligned}$$

35.

$$100 \xrightarrow[20\%]{\uparrow} 120.$$

$$\frac{\cancel{120}}{\cancel{100}} \quad \frac{\cancel{20}^{1}}{\cancel{120}_{6}} \times \cancel{100}^{16.67} \quad 40 = 40$$

$$\boxed{= 16.67.}$$

36.

$$100 \xrightarrow[20\%]{\uparrow} 120 \xrightarrow[10\%]{\downarrow} 108.$$

8%

$\frac{20}{12\cancel{0}} \times 10\cancel{0}$

~~16.67~~

$\frac{1\cancel{0}}{1\cancel{0}\cancel{0}} \times 12\cancel{0}.$

$\boxed{8\%\ \text{increase}}$

37.

CP ⇒ 100.

SP ⇒ 125.

$\frac{\cancel{20}}{\cancel{100}} \times \cancel{125} = 25\%.$

= 125 − 25%

= 100

% gain = CP

$\boxed{\therefore 0\%}$

Page

38.

$CP = 500.$

$SP = \frac{20}{1\cancel{00}} \times 5\cancel{00}$

$= 500 - 100.$

$\boxed{SP. = 400}$

39.

$100 \xrightarrow[10\%]{\uparrow se} 110. \xrightarrow[10\%.]{\downarrow se} 99$ $\boxed{(1\%. \text{ decrease})}$

$\frac{1\cancel{0}}{1\cancel{00}} \times 11\cancel{0}$

$\begin{array}{r} 110 \\ -11 \\ \hline 099 \end{array}$

40.

$200 + 20 = 220. = (40\%).$

$\begin{array}{r} 220. \\ +220 \\ \hline 440. \end{array}$

$\boxed{= 550.}$

$\frac{2\cancel{0}}{1\cancel{00}} \times 22\cancel{0}$ - 44.

$\frac{5\cancel{0}}{1\cancel{00}} \times 22\cancel{0}$ 110

41.

20% → rent.
30% → food.
10% → transport.
60%
40% → 18000.

80% → 36000
+ 9000
100% → 45000 → b.

42.
→ 100 $\xrightarrow{\uparrow}$ 30% 130 $\xrightarrow{\downarrow se}$ 91 $\boxed{\text{(9\% decrease)}}$.

$\frac{3\cancel{0}}{10\cancel{0}} \times 13\cancel{0} = 39.$

130
− 39
091

43.

10,000 → ~~20,000~~ → ~~30000~~ → ~~4000~~

10,000
↓
11000
↓
12000 $\frac{10}{1\cancel{00}} \times 100\cancel{00}$
↓
13000.

10,000
↓
11000. $\frac{10}{1\cancel{00}} \times 110\cancel{00}$
↓ 1100.
1100.
12100.
↓
→ $\boxed{13310.}$

$\frac{10}{1\cancel{00}} \times 121\cancel{00}$

1210.

44.

$\frac{15}{100} \times A = \frac{20}{100} \times B$. then A : B.

$\frac{A}{B} = \frac{\cancel{20}^{4}}{\cancel{15}_{3}} = \boxed{4:3}$

45.

→ CP → 800.

% gain → 25%.

$\frac{25}{\cancel{100}} \times 8\cancel{00} = 200$

$\boxed{\begin{aligned} SP &= 800 + 200. \\ &= 1000 \end{aligned}}$

46.

→ CP = 200.

SP = 250.

⇒ 200 ⟶ 250.

$= \frac{250 - 200}{20\cancel{0}} \times 10\cancel{0}$

$= \frac{\cancel{50}^{25}}{2}$

$\boxed{= 25\%}$

47.

→ SP = 720

% profit = 20%

$\frac{20}{720 \cdot 25} \times 100$

= 600

48. SP

SP = 15% loss

CP = 500

$\frac{15}{100} \times 500$

= 75

$\boxed{SP = 425}$

49.

→ CP = 1500.

SP = 10% ↓

$\frac{10}{1\cancel{0}\cancel{0}} \times 15\cancel{0}\cancel{0}$

1500

150.

1350

$\boxed{1350}$

50

SP = 130.

$\frac{1\cancel{0}}{1\cancel{0}\cancel{0}} \times 13\cancel{0}.$

130

13

17%

$\boxed{17\%}$